Implementation of Hybrid BWM-TOPSIS Method in the Selection of Tour Guide (Case Study: Guidemu)

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***Abstract*— Tour guides play significant and varied roles in tourism. Guidemu is an app that helps connect tourists with tour guides. A system for selecting tour guides has been implemented by Guidemu. However, the decision-making of the system still chooses a tour guide only based on the closest to the tourist destination. In this case, selecting a tour guide that relies on only one criterion is not optimal enough to decide the best tour guide based on the needs of Guidemu. Therefore, other relevant criteria must be considered in the selection of tour guides. This study proposes to implement a decision support system using Multi-Criteria Decision Making (MCDM) which provides the best alternative based on specific criteria. Of the many MCDM methods introduced, a combination of the Best-Worst Method (BWM) and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) was chosen to solve this decision problem. One of the new MCDM methods, namely BWM is used to calculate the criteria weights, while TOPSIS is used to determine the ranking of alternatives. Eight criteria were used in this study. The test results showed that the algorithm validity scored 100% by comparing the recommendations from the system with manual calculations. Furthermore, the accuracy reached 83.88% by comparing the recommendations from the system with the expert. It can be concluded that the proposed hybrid BWM-TOPSIS method is proper in the selection of Guidemu tour guides shown by good accuracy results.**

Keywords—tour guide selection, BWM, TOPSIS.

I. INTRODUCTION

Tourism has become an important industry with considerable influence on economic growth [1]. Indonesia is one of the tourist destinations in the world owing to its cultural diversity and natural resources. [2]. Based on Law Number 10 of 2009 concerning tourism, there are 13 types of tourism businesses, one of which is a tour guide. According to [3], tour guides are people who combine a high standard of enthusiasm, knowledge, personality, behavior, and ethics to guide a group of people or individuals to important places while providing interpretation and commentary. Tour guides are known to play a significant role in the tourism industry.

Guidemu is a tour guide service provider that connects tourists with local tour guides through mobile applications. The application connects tourists to the tour guides selected by the system. However, the decision-making system in Guidemu still selects tour guides based on the shortest distance from the tourist destination. In this case, selecting a tour guide based on only one criterion is no longer efficient and is less than optimal. To achieve maximum performance, other relevant criteria must be considered.

To address this issue, a study was done to build a decision support system for selecting the best tour guide based on predetermined criteria using Multi-Criteria Decision Making (MCDM). The MCDM method is used to evaluate or select a limited number of alternatives, or to select the best one from a set of alternatives [4]. This method assesses and prioritizes the alternatives that best satisfy a given set of criteria. Criteria can be measures, rules, or standards used in decision-making [5].

The results of the decision-making on the selection of tour guides play an important role in achieving the goals of Guidemu in providing the best decisions not only for tourists but also for tour guides. Many decision-making problems have been solved using the MCDM approach, but very few studies have focused on the selection of tour guides. Research [6] has succeeded in building a decision-making system for the selection of tour guides using the MCDM method, namely simple additive weighting (SAW). However, the process of determining the criteria to the weighting of the criteria was not explained further, in addition, there was no expert involved in the decision-making process and any form of testing was not carried out on the proposed method. Therefore, this study aims to comprehensively explain the process of developing a decision-making system that focuses on the selection of tour guides in Guidemu.

Currently, several methods can be used to solve the MCDM problems. The technique for order of preference by similarity to ideal solution (TOPSIS) was selected because it has proven useful in solving many real-world MCDM problems and has been regarded as one of the primary methods in the decision-making process [7]. However, TOPSIS can only be used for criteria whose weights are known or calculated in advance [8].

However, determining criteria weights is a key element in MCDM problems. The weight of the criteria plays an important role in measuring the overall alternative preference [9]. Therefore, the recently introduced MCDM method, the best-worst method (BWM), was chosen because it can identify the optimal weight of a set of criteria based on the preferences of a single decision maker [10]. Furthermore, compared with similar weighting methods, this method is easier to apply and produces a significantly more consistent weighting result [11].

Consequently, the hybrid BWM-TOPSIS method was proposed for implementation in the tour guide selection system of Guidemu.

II. THEORETICAL FRAMEWORK

A. Multiple-Criteria Decision Making

MCDM is a decision-making method used to determine the best alternative from a set of alternatives based on certain criteria determined by a decision maker [5]. MCDM is one of the most widely used decision-making methods. The MCDM process follows a common working principle [12]:

1. Identifying the goal.
2. Selection of criteria.
3. Selection of alternatives.
4. Selection of weighing method.
5. Method of aggregation.

B. Technique for Order of Preference by Similarity to Ideal Solution

TOPSIS is an MCDM method that was first introduced by Hwang and Yoon in 1981. This method is implemented based on the concept that the best alternative chosen not only has the shortest distance from the positive ideal solution but also has the longest distance from the negative ideal solution. A positive ideal solution maximizes the benefit criterion and minimizes the cost criterion, whereas a negative ideal solution maximizes the cost criterion and minimizes the benefit criterion. TOPSIS is widely used to address practical decision-making problems because of its simplicity, ease of understanding, efficient computing, and ability to calculate the relative performance of alternative decisions in a simple mathematical form [13].

Implementation of TOPSIS method in research [14] has helped an institution in the selection of scholarship applicants with the aim of replacing the manual selection process. The results of the method show the ranking of applicants based on predetermined criteria and weights.

The TOPSIS procedure consists of the following steps [5]:

1. Assign alternative (i) and criteria (j) into a decision matrix.

$$\begin{bmatrix} x_{11} & \cdots & x_{1j} \\ \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} \end{bmatrix}$$

Where: $i = 1,2,3,\dots,m$ and $j = 1,2,3,\dots,n$

2. Determine the normalized decision matrix. The value of each matrix element is calculated using the following formula:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}} \quad (1)$$

3. Determine the weighted normalized decision matrix. With weight $w = (w_1, w_2, \cdots, w_j)$, the value of each matrix element is calculated using the following formula:

$$\mathrm{y_{ij}} = \mathrm{w_j} \cdot \mathrm{r_{ij}} \quad (2)$$

4. Determine the matrix of positive ideal solution (A^+) and the matrix of negative ideal solution (A^-).

$$\begin{aligned} \mathrm{A}^+ &= [y_1^+ \quad y_2^+ \quad \cdots \quad y_j^+] \\ A^- &= [y_1^- \quad y_2^- \quad \cdots \quad y_j^-] \end{aligned} \quad (3)$$

Where:

$$y_j^+ = \begin{cases} max_i \, y_{ij}, & \text{if j is benefit attribute} \\ min_i \, y_{ij}, & \text{if j is cost attribute} \end{cases}$$

$$y_j^- = \begin{cases} min_i \, y_{ij}, & \text{if j is benefit attribute} \\ max_i \, y_{ij}, & \text{if j is cost attribute} \end{cases}$$

5. Calculate the alternative distance to the matrix of the positive ideal solution (D^+) and the alternative distance to the matrix of the negative ideal solution (D^-).

$$D_i^+ = \sqrt{\sum_{i=1}^{n} \left(y_{ij} - y_i^+\right)^2} \quad D_i^- = \sqrt{\sum_{i=1}^{n} \left(y_{ij} - y_i^-\right)^2} \quad (4)$$

6. Determine the preference value (V) for each alternative using the formula:

$$V_i = \frac{D_i^-}{D_i^- + D_i^+} \quad (5)$$

C. Best-Worst Method

BWM is a new MCDM method developed by Dr. Jafar Rezaei in 2015. BWM is based on a pairwise comparison of the best and worst criteria to determine the weight of the criteria in the MCDM problem [15]. The best criteria are the most desirable or most important criteria among other criteria, whereas the worst criteria are the least desirable or least important criteria among other criteria. BWM overcomes the drawbacks of similar methods, such as inconsistency, as it significantly reduces the number of comparisons required compared with other pairwise comparison-based methods [16]. By eliminating secondary comparisons, this method proved to be much more efficient and easier for determining weights [17]. Statistical findings also show that BWM requires fewer comparison data but produces more consistent comparisons [18].

Research [11] used the hybrid BWM-TOPSIS method to evaluate operational performance of a power grid enterprise in China from a sustainability perspective. BWM was employed to determine the weights of all criteria, and TOPSIS was applied to rank the operation performance of a power grid enterprise. The proposed method assessed effective and practical in performance evaluation.

The steps of the BWM calculation are as follows [15]:

1. Determine a set of decision criteria.
2. Determine the best and the worst criteria.
3. Determine the preference of the best criteria over all other criteria by using a number between one and nine.
4. Determine the preference of all criteria over the worst criteria by using a number between one and nine.
5. Find the optimal weights.

III. Research Method

The research stages of the decision support system for tour guide selection in Guidemu using the hybrid BWM-TOPSIS method are shown in Figure 1.

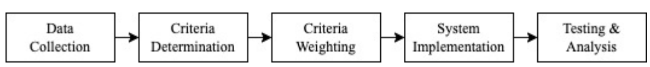

Fig. 1. Research Stages

A. Data Collection

Data collection was done through observation and interview.

1) Observation

Observations were conducted to obtain the data required for the research, such as registered tour guide data and their attributes, history of tour guide selection, as well as algorithms and criteria used in the process of selecting a tour guide for Guidemu.

2) Interview

The interview involved one expert who was the main decision-maker in Guidemu. The interviews revealed an overview of the company profile, company goals, characteristics of registered tour guides and the parts involved, obstacles to the tour guide selection system, and expectations for the development of further decision-support systems. The interview sessions also helped identify the criteria and determine their weights.

B. Criteria Determination

This stage aims to determine the relevant criteria for the tour guide selection process using MCDM by referring to literature review and field research. The literature review identified the criteria based on previous research. Meanwhile, field research identified criteria based on a company's goals and business needs.

C. Criteria Weighting

At this stage, the weight of the predetermined criteria was calculated using the BWM by collecting data from the expert. The BWM calculation to determine the optimal weight was performed in Microsoft Excel using BWM Solver [19]. The output of this stage is the weight for each criterion along with the consistency ratio used to assess the level of consistency of the weight given by the decision-maker. This weight measures the relative importance of the criteria, the greater the weight of a criterion, the more important it is compared with other criteria [20].

D. System Implementation

This stage was conducted by implementing TOPSIS to rank alternative tour guides based on the weights of the criteria calculated using BWM with the final result providing recommendations based on the best one.

E. Testing & Analysis

This stage tested the implementation of the hybrid BWM-TOPSIS method on the system and analyzed the test results. This section consists of an algorithm validation test and accuracy test.

IV. Result and Discussion

A. Result of Criteria Determination

Based on [14], three stages are required to determine the decision-making criteria using the TOPSIS. The results for each stage are as follows:

1) Criteria Used

Five criteria were obtained from the literature review, and seven from the field research. As a result, there were eight unique criteria used in this study which are shown in Table 1.

TABLE 1. CRITERIA USED

Criteria	Description	Source	
		Literature Review	*Field Research*
C_1	Length of work experience (year)	✓	
C_2	Number of work areas	✓	
C_3	Number of reputations	✓	✓
C_4	Number of languages	✓	✓
C_5	Number of skills	✓	✓
C_6	Distance to destination (kilometer)		✓
C_7	Number of orders		✓
C_8	Commitment level		✓

The literature review was conducted on several previous research [6], [21], [22]. However, research [6] was chosen as the main reference because it has the most related research topics among other references. The field research was conducted through interviews with the expert to propose criteria based on the needs and objectives of Guidemu which not only prioritizes tour guides with the best talents but also eliminated the culture of seniority by providing opportunities for novice tour guides to be selected in order to achieve equity.

2) Weighting Rule

A weighting rule for each criterion was required to normalize the raw data obtained from each alternative transformed into a weighted value on a scale of one to five, where one was the lowest value and five was the highest value. The weighting rules are presented in Table 2.

TABLE 2. WEIGHTING RULES

Criteria	Raw Data	Weight
C_1	< 2	1
	2 – 4	2
	5 – 7	3
	8 – 10	4
	> 10	5
C_2	< 2	1
	2 – 4	2
	5 – 7	3
	8 – 10	4
	> 10	5
C_3	< 16	1
	16 – 30	2
	31 – 45	3
	46 – 60	4
	> 60	5
C_4	< 2	1
	2 – 3	2
	4 – 5	3
	6 – 7	4
	> 7	5

C_5	< 2	1
	2 – 3	2
	4 – 5	3
	6 – 7	4
	> 7	5
C_6	< 2	1
	2 – 4	2
	5 – 7	3
	8 – 10	4
	> 10	5
C_7	< 16	1
	16 – 30	2
	31 – 45	3
	46 – 60	4
	> 60	5
C_8	Very Low	1
	Low	2
	Normal	3
	High	4
	Very High	5

3) *Criteria Attribute*

The attribute criterion comprises benefits and costs, where the benefit attribute prioritizes the highest value, and the cost attribute prioritizes the lowest value. Table 3 shows the attributes of each criterion.

TABLE 3. CRITERIA ATTRIBUTE

Criteria	**Attribute**
C1	Benefit
C2	Benefit
C3	Benefit
C4	Benefit
C5	Benefit
C6	Cost
C7	Cost
C8	Benefit

B. *Result of Criteria Weighting*

After determining the criteria, they were weighted using the BWM. Data collected from expert is presented in Table 4, Table 5, and Table 6.

TABLE 4. BEST AND WORST CRITERIA

Best Criteria	**Worst Criteria**
C_5	C_6

TABLE 5. PREFERENCES OF BEST-TO-OTHERS CRITERIA

Best-to-Others	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
C_5	2	5	1	2	1	9	5	3

TABLE 6. PREFERENCES OF OTHERS-TO-WORST CRITERIA

Others-to-Worst	C_6
C_1	7
C_2	3
C_3	7
C_4	7
C_5	7
C_6	1
C_7	3
C_8	7

After obtaining the required data, BWM calculation was performed using Microsoft Excel. The weighting results are listed in Table 7.

TABLE 7. BWM CALCULATION RESULTS

Criteria	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
Weight	0.15	0.06	0.23	0.15	0.23	0.02	0.06	0.10
CR	0.07							

The consistency ratio (CR) was used to verify the reliability of pairwise comparison results [23]. A CR value smaller than or close to zero indicates higher consistency, whereas a CR value higher than or close to one indicates lower consistency during pairwise comparisons [15]. Therefore, the value of 0.07 obtained was considered very consistent (<0.10). Therefore, it can be concluded that the weighting results were valid and could be used.

C. *Result of System Implementation*

TOPSIS was implemented in the system to provide tour guide recommendations. Three tour guide data samples were used, as listed in Table 8.

TABLE 8. TOUR GUIDE SAMPLE

Alternative	**Criteria**							
	C1	***C2***	***C3***	***C4***	***C5***	***C6***	***C7***	***C8***
Hidayat	5	2	21	4	4	8	6	High
Haryanto	10	9	18	2	5	3	62	Normal
Santoso	5	7	13	4	4	6	73	Very High

The stages below demonstrate how the system determines tour guide recommendations.:

(1) The decision matrix (X) was created by converting the raw tour guide data into weighted values according to predetermined weighting rules, as listed in Table 9.

TABLE 9. VALUE OF MATRIX X

3	2	2	3	3	5	1	4
4	5	2	2	3	5	5	3
3	4	1	3	3	5	5	5

(2) The normalized decision matrix (R) was determined. Based on Equation (1), the matrix R is presented in Table 10.

TABLE 10. VALUE OF MATRIX R

0.514	0.298	0.667	0.640	0.577	0.577	0.140	0.566
0.686	0.745	0.667	0.426	0.577	0.577	0.700	0.424
0.514	0.596	0.333	0.640	0.577	0.577	0.700	0.707

(3) The weighted normalized decision matrix (Y) was determined. Based on Equation (2) with $\mathrm{w} = (0.15, 0.06, 0.23, 0.15, 0.23, 0.02, 0.06, 0.10)$. Thus, matrix Y is presented in Table 11.

TABLE 11. VALUE OF MATRIX Y

0.077	0.018	0.153	0.096	0.133	0.012	0.008	0.057
0.103	0.045	0.153	0.064	0.133	0.012	0.042	0.042
0.077	0.036	0.077	0.096	0.133	0.012	0.042	0.071

(4) The matrices of the positive ideal solution (A^+) and the negative ideal solution (A^-) were determined. Based on Equation (3), the following results were obtained, as listed in Table 12.

TABLE 12. IDEAL SOLUTION

Criteria	A^+	A^-
C_1	0.103	0.077
C_2	0.045	0.018
C_3	0.153	0.077
C_4	0.096	0.064
C_5	0.133	0.133
C_6	0.012	0.012
C_7	0.008	0.042
C_8	0.071	0.042

(5) The distance between the values of each alternative with A^+ denoted by D^+ and the distance between the values of each alternative with A^- denoted by D^- were determined. The results based on Equation (4) are presented in Table 13.

TABLE 13. ALTERNATIVE IDEAL SOLUTION

Alternative	D^+	D^-
Hidayat	0.040	0.091
Haryanto	0.054	0.085
Santoso	0.088	0.046

(6) The preference value (V) was determined by calculating the closeness of each alternative to the ideal solution using Equation (5). The results and alternative ranks are presented in Table 14.

TABLE 14. ALTERNATIVE RANK AND PREFERENCE SCORE

Alternative	V	Rank
Hidayat	0.695	1
Haryanto	0.611	2
Santoso	0.345	3

(7) After ranking each alternative, the best tour guide is selected based on the highest preference value. The tour guide recommended by the system was Hidayat with a score of 0.695.

D. *Result of Testing & Analysis*

Twelve tests were carried out with three different tour guides used for each test.

1) Algorithm Validation Testing

This test was conducted by comparing the results of the recommendations from the system with the results of manual calculations to confirm whether the system functioned properly and correctly.

TABLE 15. ALGORITHM VALIDATION TESTING

Number	Recommendation Results		Valid
	System	*Manual*	
1	Hidayat	Hidayat	✓
2	Kurniawan	Kurniawan	✓
3	Sitompul	Sitompul	✓
4	Laksmiwati	Laksmiwati	✓
5	Hariyah	Hariyah	✓
6	Usada	Usada	✓
7	Usamah	Usamah	✓
8	Putra	Putra	✓
9	Puspasari	Puspasari	✓
10	Mardhiyah	Mardhiyah	✓
11	Wirawan	Wirawan	✓
12	Santoso	Santoso	✓

Table 15 shows that the data obtained from the algorithm validation testing are valid, that is, the recommendation results from the system for manual calculations are similar. Therefore, it can be concluded that the system has been implemented properly and correctly because it has an algorithm validity of 100%.

2) Accuracy Test

To measure the accuracy of the proposed method, an accuracy test was conducted by comparing the results of the recommendations from the system with the results of expert analysis.

TABLE 16. ACCURACY TEST RESULT

Number	Recommendation Results		Valid
	System	*Expert*	
1	Hidayat	Rahman	
2	Kurniawan	Kurniawan	✓
3	Sitompul	Sitompul	✓
4	Laksmiwati	Laksmiwati	✓
5	Hariyah	Hariyah	✓
6	Usada	Usada	✓
7	Usamah	Usamah	✓
8	Putra	Firmansyah	
9	Puspasari	Puspasari	✓
10	Mardhiyah	Mardhiyah	✓
11	Wirawan	Wirawan	✓
12	Santoso	Santoso	✓

From Table 16, there are ten valid test data and two invalid test data. Accuracy is calculated by dividing the number of valid tests by the total number of tests. Thus, the accuracy of the proposed method was 83.33%, which was classified as good accuracy [24]. Meanwhile, the existing system that based on only one criterion, namely the shortest distance, obtained five out of twelve valid test data, so that the existing system has an accuracy of 41.66%.

E. Discussion of Results

Our results complement previous research [6] in the process of implementing a decision-making system in the selection of tour guides, especially at the stage of criteria identification, criteria weight calculation, algorithm validation, and accuracy test. The accuracy test results found a significant increase of 100% between the existing system and the proposed system, from 41.66% to 83.33% accuracy.

V. CONCLUSION

Eight criteria were applied in this study, namely length of work experience, number of work areas, number of reputations, number of languages, number of skills, distance to destination, number of orders, and commitment level. The proposed BWM aids in establishing the criteria weights in a simple and reliable manner. Meanwhile, TOPSIS, which has been applied, has proven successful in determining tour guide recommendations based on predetermined criteria and weights. The results of the algorithm validation test indicated that the system was implemented properly and correctly, as indicated by a validity value of 100%. The accuracy test showed that the proposed method had an accuracy of 83.33%. Overall, the test results demonstrated that the hybrid BWM-TOPSIS method can be applied to a decision support system for tour guide selection on Guidemu.

However, this study has several limitations. First, because Guidemu is start-up company and has a lean organizational structure, only one expert can be involved in this study. Second, due to time constraints, only twelve data samples were used for testing purpose, the use of different data samples affects the accuracy of the results.

Future research should use a combination of other MCDM methods to achieve a higher accuracy. Given that BWM is a relatively new MCDM method, it would be preferable to employ a combination of these methods in different case studies. Other testing approaches such as sensitivity analysis can be added to enhance the test process.

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